

WHAT DOES RETRIEVED MEMORY BUY? ACCURACY, EVIDENCE AND WRITE COST

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ABSTRACT

Agent memory can reduce information loss under a restricted prompt budget, but query-only accounting may conceal its construction cost, and a plausible ranking heuristic may not survive a controlled comparison against a simple lexical baseline. We report two complementary studies of memory selection in JiuwenSwarm. The first is an operational comparison of file-prefix memory and retrieved working memory under matched full-section character budgets. On 20 adapted single-session questions, retrieved memory raises answer accuracy from 0.0% to 65.0% at a restricted budget and from 60.0% to 95.0% at an expanded budget, but model-backed construction makes it 21.70 and 15.64 times more expensive per question. The second study isolates selector behaviour on a fixed-hash stratified LongMemEval-S subset of 84 memory episodes under a frozen candidate bank, a fixed whole-item packing policy and 512, 1024 and 2048 pinned-token budgets. There the outcome reverses: the deployed hybrid scorer is 36.1, 29.4 and 33.7 percentage points *worse* than a BM25 baseline, and removing only its temporal term recovers 27.0 points at the 1024-token budget. A score-level analysis identifies the mechanism: a fixed, query-independent freshness weight can outweigh a positive lexical relevance gap, displacing relevant older history. We retain the negative result and report its scope, including a full-history retrieval sweep over 500 histories and component controls that separate representation, acquisition and ranking effects. The two studies together show that a memory benefit observed under a truncated-file baseline does not license a claim that the hybrid ranking itself is the better selector, and that construction cost and selector quality must be measured separately.

1 INTRODUCTION

A language-model agent can retain project facts in a file and expose a prefix of that file to its next question. This is inexpensive to construct, but a fixed prompt allowance can remove relevant facts before the model sees them. An alternative is to store exchanges as separate memory items and select a subset for each question. That design introduces a second resource decision: how the items are constructed. If construction itself invokes a model, measuring only the final query can give an incomplete account of operational cost. A useful comparison therefore needs to follow the information delivered to the query and the model calls required to prepare it.

It also introduces a methodological hazard. When a retrieval system is compared only against a deliberately weak baseline, a favourable result is easy to obtain and easy to over-read. Replacing relevant items with a truncation artefact can make almost any selector look effective, and this kind of comparison says more about the baseline than about the selector. We therefore report two studies whose designs make different sacrifices. The first keeps the real operational pairing and accepts that construction and presentation differ jointly. The second freezes everything except the ordering rule, which costs the realism of the operational system but makes the selector comparison interpretable.

This paper presents an auditable systems study of that trade-off together with a controlled follow-up. The research question is: how do file-based project memory and hybrid retrieval compare in answer accuracy and total model cost, and does the hybrid ranking rule itself select better memory than a standard lexical baseline? Four sub-questions structure the investigation. RQ1 asks whether accuracy differs between the two operational arms when both receive the same character budget.

RQ2 asks whether complete evidence recall is associated with correct answers, and whether partial evidence can suffice. RQ3 asks how total token cost, including memory-write calls, compares between the arms. RQ4 asks whether the hybrid ranking rule outperforms a lexical baseline when representation, packing policy and token budget are held fixed.

The answers are not uniform. RQ1 through RQ3 describe a real operational advantage in the audited implementation pair: retrieved memory delivers more complete evidence and more accepted answers, at a substantially higher total cost. RQ4 returns the opposite verdict. Under a frozen candidate bank and a fixed packing policy, the deployed hybrid scorer is markedly worse than BM25, and ablating only its temporal term recovers most of the loss. We report both outcomes, because the second is what a reader needs in order to interpret the first. A memory benefit observed against a truncated-file baseline is a statement about that baseline as much as about the selector.

The study audits an existing implementation rather than proposing a new retrieval algorithm. In the operational comparison both arms receive identical facts, questions, model identity, and system prompt; only the memory-construction and selection protocol differ. The baseline writes facts to an isolated `JIUWENSWARM.md` file and presents a prefix-truncated view. The enhanced arm writes completed user-assistant exchanges to process-local L1 memory through model-backed conversations and selects items using a fixed lexical ranking. Neither arm employs learned embeddings, reflection modules, or external tools. In the controlled comparison the candidate bank, metadata, memory wrapper, token cap and first-fit overflow policy are shared across all selectors; only the scoring rule changes.

The study contributes an auditable comparison, a cost decomposition, a controlled selector evaluation with a reported negative result, and a question-level diagnostic record. The comparison pairs identical questions under restricted and expanded memory budgets instead of treating the two budgets as independent datasets. The decomposition separates query tokens from model-backed writes, allowing an accuracy advantage to be assessed alongside its construction expense. The controlled evaluation replaces a single generous comparison with six selectors across three budgets, with paired cluster-bootstrap intervals and Holm-corrected tests, and it isolates the contribution of the temporal term. The diagnostics distinguish missing complete evidence from an incorrect answer and expose a case where a partial fact still contains an accepted answer. These contributions concern measurement and implementation behaviour; they do not establish a new retrieval algorithm or state-of-the-art performance.

The scope is deliberately narrow. The experiments do not validate an end-to-end paper-generation pipeline, do not test multi-session reasoning or temporal updates inside the frozen bank, and do not compare against published memory systems as systems. The adapted LongMemEval slice used in the operational study covers only single-session factual extraction; the controlled study broadens category coverage but still evaluates a question-answering component rather than a deployed product. All reported intervals are descriptive or cluster-bootstrap intervals over the stated question sets, and we mark explicitly where a single-run protocol does not support a significance claim.

1.1 SUMMARY OF FINDINGS

We state the headline results plainly so that the remainder of the paper can be read as support rather than as a surprise.

- **Retrieved memory helps against a truncated file at matched character budgets.** Accuracy rises from 0.0% to 65.0% at 600 characters and from 60.0% to 95.0% at 2400 characters, with zero baseline-only successes at either budget.
- **Its construction cost dominates its operating cost.** Including model-backed writes, retrieved memory costs 21.70 times the baseline per question at the restricted budget and 15.64 times at the expanded budget, even though its query-token totals are lower.
- **The hybrid controller does not select better than BM25.** Under a frozen bank it is 36.1, 29.4 and 33.7 points worse at 512, 1024 and 2048 tokens, a result that survives Holm correction at every budget.
- **The temporal term causes most of the deficit.** Removing only the time-decay term from the same scorer recovers 27.0 points at 1024 tokens while leaving source records, lexical scoring and packing unchanged.

- **Evidence delivery tracks correctness.** Selector ranking quality is visible in both answer accuracy and the fraction of annotated evidence turns that reach the prompt, and the two orderings agree across all six selectors.

The remainder of the paper is organised as follows. Section 1 through Section 3 set out the protocol and the audited implementation. Section 4 defines both experimental designs and their statistical treatment. Section 5 reports the operational study, Section 6 the controlled selector study, and Section 7 the score-level mechanism behind the negative result. Section 8 discusses what the two designs jointly license and what they do not.

2 RELATED WORK

2.1 AGENT MEMORY SYSTEMS

Recent surveys argue that agent memory is evolving into a full-fledged data management layer that governs storage, retrieval, consolidation, and lifecycle policies throughout execution (Zhou et al., 2026). Complementing this view, enterprise-grade substrates emphasize durable, organization-wide retention of task state and procedural knowledge for long-horizon workflows (Alake et al., 2026). At the mechanism level, memory-as-a-tool approaches convert transient critiques into retrievable guidelines to amortize inference-time reasoning (Gallego, 2026). Unified episodic–semantic designs further explore spreading-activation retrieval to connect long-term facts with task-specific episodes (Jiang et al., 2026).

2.2 RETRIEVAL METHODS FOR AGENT MEMORY

Neural ranking and retrieval provide the algorithmic backbone for scoring and selecting relevant records under diverse evidence sources (Trabelsi et al., 2021). Hybrid retrievers that fuse vector semantic search with lexical signals have demonstrated accuracy gains over single-source methods (Sawarkar et al., 2024; Steffens, 2026). Beyond query–content similarity, longitudinal studies highlight the importance of temporal dynamics in retrieval effectiveness (Ovcharov, 2026). Our setting adopts a hybrid scoring paradigm tailored to agent memory by integrating semantic similarity with temporal decay, access frequency, and importance weighting.

2.3 MULTI-AGENT COLLABORATION

Multi-agent evaluation frameworks underscore the benefits and challenges of coordinating several LLM agents to assess capabilities and systemic risks (Rasheed et al., 2024). Domain-specific, hierarchical teams showcase how role specialization and delegated reasoning can drive autonomous scientific discovery (Rothfarb et al., 2025). From a systems perspective, organization-scale memory infrastructures are advocated to support coherent collaboration across extended processes (Alake et al., 2026). Nevertheless, concrete protocols for synchronizing, reconciling, and governing shared memories across teams remain under-specified.

2.4 LLM-BASED AUTONOMOUS AGENTS

LLM-based agents increasingly rely on tool use and iterative interaction to execute complex tasks, yet alignment and finetuning must bridge the gap between generation and effective action (Wang et al., 2024). Surveys of advanced LLM capabilities highlight persistent issues—temporal knowledge gaps, arithmetic robustness, and hallucination—that motivate external memory and retrieval (Roffo, 2024). As agents accumulate long-term histories, privacy analyses reveal new attack surfaces in memory stores, calling for principled management and access policies (Chen et al., 2026). In parallel, techniques that distill feedback into persistent, retrievable rules aim to stabilize behavior across episodes (Gallego, 2026).

Building on calls for agent-native memory infrastructures, we introduce a three-level working–task–project architecture with explicit per-layer TTL and capacity controls, enabling lifecycle governance that is not addressed by existing designs (Zhou et al., 2026). Departing from standard hybrid retrievers, our scoring function couples semantic similarity with temporal decay, access frequency, and importance weighting to prioritize salient, timely, and reused items (Sawarkar et al.,

2024; Ovcharov, 2026). Finally, unlike prior multi-agent efforts that emphasize evaluation or hierarchical role decomposition, we formalize a team memory synchronization protocol and validate the full stack in JiuwenSwarm (Rasheed et al., 2024; Rothfarb et al., 2025).

3 METHOD

The method diagram contrasts the two executed paths from a shared fact episode to a fresh query conversation and a paired audit. The baseline constructs a file-derived section and retains its prefix. The retrieved arm constructs dialogue-shaped memory items, ranks them, and packs complete formatted items into the allowed section. The same model answers both queries, but construction and presentation differ jointly. Accordingly, the estimand is the observed difference between these operational systems under a matched section budget, rather than the isolated effect of lexical retrieval. The later controlled study is designed precisely to separate the two, at the cost of the operational realism preserved here.

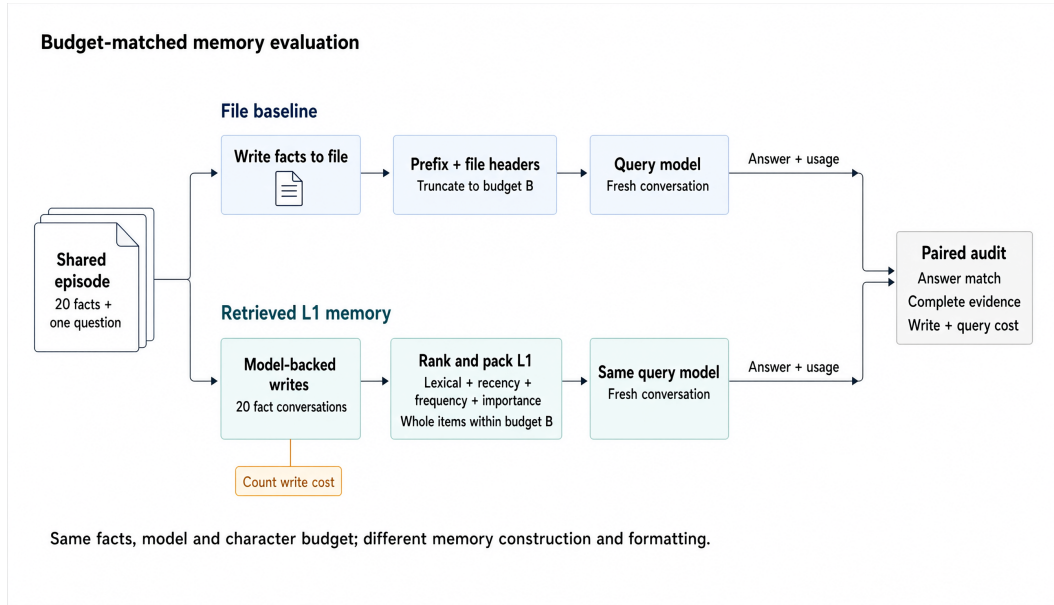


Figure 1: Budget-matched evaluation of two operational memory paths. The retrieved arm incurs model-backed write cost before its isolated query. Equal character caps include each arm’s own formatting overhead. This schematic illustrates the audited implementation; it is not an experimental result.

3.1 FILE-BASED PROJECT MEMORY

The baseline uses ProjectMemoryRail. Facts are written directly to an isolated JI-UWENSWARM.md file without model-backed seeding. At query time, the runner reads the file, includes its filename and modification metadata, and truncates the complete memory section to the allowed prefix. All of these characters count against the budget. There is no query-dependent re-ordering: later facts can be removed regardless of relevance, and the truncation boundary may split a fact. Consequently, the arm is a concrete file-prefix baseline rather than an embedding retriever. If complete evidence is absent, that establishes what the model did not receive under this protocol; it does not establish how a differently ordered or reformatted file would perform.

3.2 HYBRID SELECTION FROM WORKING MEMORY

The enhanced arm uses EnhancedMemoryRail to store completed user-assistant exchanges in process-local L1 memory. Each fact is supplied through a separate model-backed conversation, so the resulting item contains both the supplied text and an assistant response. Retrieval combines

Jaccard overlap of token sets with recency, access frequency and a heuristic importance score. The formalization below gives the audited default weights and normalization; these coefficients were not fitted on the evaluation questions. A fixed-size candidate set is ranked, each display item is individually capped, and a greedy packer adds complete formatted items while space remains. An item that does not fit is skipped, allowing a later shorter item to be considered. Selection and packing can therefore omit a fact for different reasons, even when total memory capacity is sufficient. This path invokes no embedding model and provides no cross-restart persistence. The implementation defines other memory layers, but they are not exercised by this Rail. MemGPT supplies conceptual context for tiered-memory terminology (Packer et al., 2023); its autonomous context controller is not implemented here.

3.3 ISOLATION AND COST ACCOUNTING

Each public-data question receives its own fact episode and fresh agent instances. For the baseline, facts are written directly to disk. For the enhanced arm, the runner conducts model-backed memory-writing conversations: each fact episode is presented as a user turn, the model produces an assistant acknowledgement, and the completed exchange is stored in L1. This seeding phase requires 400 model calls per budget and is the source of the write-cost asymmetry. The formal question then uses a new conversation ID so that no seeding dialogue leaks into the query context. The runner verifies isolated query history and confirms actual memory injection before scoring. Arm execution order is alternated across questions to control for potential API-level rate-limiting or temporal drift during the run; because agents are instantiated fresh per question, there is no cross-question shared state contamination to mitigate. Both query-time token usage and memory-write token usage are recorded separately. Latency measurements include the write phase and are amortised over the paired observations. Correctness is assessed by accepted-answer string matching against a predefined answer list; evidence recall requires the complete target fact to survive prompt truncation and appear in the injected context. The evaluation design adapts the single-session extraction paradigm of LongMemEval (Wu et al., 2024), though the resulting slice is not the official benchmark.

3.4 BUDGET AND COST FORMALIZATION

Let q be a question, M its episode memory, and B the cap on the entire injected memory section, including headings and metadata. The baseline constructs a file-derived section and keeps its prefix. The retrieved arm scores stored exchanges, takes at most $k = 10$ candidates, and greedily appends complete formatted items that fit. An over-budget item is skipped rather than partially appended; later smaller items can still be considered. Thus equal B does not imply equal usable fact content or identical prompt formatting.

$$s(q, m) = 0.4J(T(q), T(m)) + 0.32^{-a(m)/h} + 0.2 \min\left\{\frac{\log(1 + f(m))}{\log(101)}, 1\right\} + 0.1I(m), \quad (1)$$

where T lowercases text and keeps alphanumeric/underscore word tokens longer than one character; $J(A, C) = |A \cap C|/|A \cup C|$ (zero for an empty set); $a(m)$ is age in seconds; $h = 86400$ is the decay half-life; $f(m)$ is the recorded access count; and $I(m)$ is a bounded heuristic importance score. These are the implementation’s default weights, not learned or tuned coefficients. No embedding model is invoked. L1 holds at most 20 exchanges with a one-hour time-to-live; individual displayed exchanges are capped at 500 characters before packing.

The cost estimand includes initialization through model-backed memory writes:

$$C_g = \frac{\sum_{r \in \mathcal{Q}_g} u_r + \sum_{w \in \mathcal{W}_g} u_w}{N_{\text{valid}}}, \quad L_g = \frac{\sum_{r \in \mathcal{Q}_g} t_r + \sum_{w \in \mathcal{W}_g} t_w}{N_{\text{valid}}}. \quad (2)$$

Here u and t denote recorded token usage and elapsed time, respectively; $\mathcal{Q}_g$ and $\mathcal{W}_g$ include all recorded query and write attempts. The denominator counts valid matched question pairs. These timing sums exclude unrecorded process initialization and are not a full application wall-clock time. Missing usage remains unknown rather than zero.

For each valid pair, let $y_{g,i}$ indicate an accepted-answer substring match and $e_{g,i}$ indicate that the complete target fact survived in the injected text:

$$A_g = \frac{1}{N} \sum_i y_{g,i}, \quad R_g = \frac{1}{N} \sum_i e_{g,i}, \quad \Delta A = A_{\text{retrieved}} - A_{\text{file}}. \quad (3)$$

The two indicators differ: a truncated fact can contain a sufficient answer span. Wilson intervals describe finite-question uncertainty only. They do not estimate model-run variability, correct the selected slice’s bias, or establish significance.

Table 1: Audited execution protocol. No component ablation or weight tuning is claimed.

Step	Operation
Prepare	Create fresh isolated agents; give both arms the same fact episode.
Construct	Write the baseline file; seed retrieved L1 through separate fact conversations.
Query	Use a fresh conversation, no tools, one model iteration, and the assigned section budget.
Audit	Check actual prompt injection and isolated history before accepting a pair.
Aggregate	Match question IDs; count writes once per run; preserve excluded attempts in costs.

3.5 CONTROLLED SELECTOR DESIGN

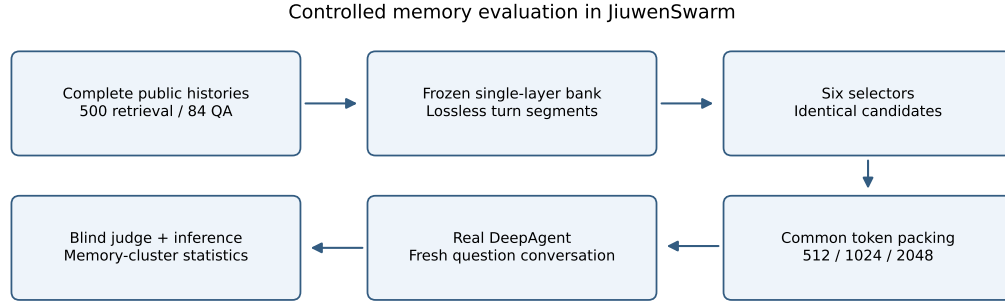
The operational comparison changes construction, representation and ordering together, so it cannot attribute its outcome to the ranking rule. The controlled study removes that ambiguity by holding everything except the ordering rule fixed. All selectors draw from the same frozen candidate bank: complete source sessions with user and assistant turns, timestamps and gold answers are retained, and each turn is split losslessly into segments bounded by 256 pinned Qwen3 BPE tokens. The bank, the item metadata, the memory wrapper and the whole-item first-fit overflow policy are identical across selectors, as are the token budgets. Only the scoring function that orders candidates changes.

Every selector instantiates the same weighted family

$$s_{\text{sel}}(q, m) = \bar{w}_s J(T(q), T(m)) + \bar{w}_t D(m) + \bar{w}_f F(m) + \bar{w}_i I(m), \quad (4)$$

where $D(m) = 2^{-a(m)/h}$ is the temporal decay, $F(m) = \min\{\log(1 + f(m))/\log(101), 1\}$ is the bounded frequency term, and $I(m)$ is the heuristic importance score. The deployed hybrid uses $(\bar{w}_s, \bar{w}_t, \bar{w}_f, \bar{w}_i) = (0.4, 0.3, 0.2, 0.1)$. The controlled variants set individual terms to zero while leaving the candidate bank, the lexical tokenizer and the packing rule untouched: *prefix* uses bank order with no scoring, *recency* keeps the temporal term alone, *jaccard* keeps lexical overlap alone, and *hybrid_no_time* removes only the temporal term and sums the remaining weighted terms directly, which preserves lexical tie order. A further *bm25* selector replaces Jaccard with a standard probabilistic lexical score using $k_1 = 1.2$, $b = 0.75$ and positive inverse document frequency (Robertson & Zaragoza, 2009). Because prefix ordering is a documented failure mode rather than a competitive baseline, and because BM25 is a strong, well understood lexical ranker, the primary comparison of the controlled study is hybrid against BM25. These are controlled selectors, not a novel learned retrieval algorithm.

Separating ranking from acquisition also separates two cost questions. The controlled study measures ranking quality at a fixed representation and therefore excludes model-backed construction from its accuracy comparisons. Whether an application pays for model-backed construction is a property of the deployment, not of the selector, and the two studies are reported separately for that reason. Section 4 gives the sampling, budgets and statistical treatment of both designs.



Micro: representation OR acquisition changed alone. Reuse: one build, 20 distinct queries.

Figure 2: Controlled evaluation protocol. A single frozen candidate bank per memory episode is served to every selector under identical token caps and an identical whole-item first-fit packing policy. Only the ordering rule varies between arms, so score differences are attributable to ranking rather than to representation or acquisition. The diagram describes the executed protocol; it reports no outcome.

4 EXPERIMENTAL SETUP

We report two studies with complementary designs. The operational comparison keeps the real system pairing and accepts that construction and presentation differ jointly. The controlled selector study freezes everything except the ordering rule and therefore isolates ranking, at the cost of the operational realism that the first study provides. The two must be read together: an advantage in the first does not imply that the ranking rule is the cause, and a ranking result in the second does not by itself describe the deployed system. Section 4.1 defines the operational comparison, Section 4.4 the controlled study, and Section 4.6 the statistical treatment shared by both.

4.1 OPERATIONAL COMPARISON: DESIGN

The evaluation uses a selected LongMemEval-S single-session-user slice converted to fixed episodes containing one target answer turn and distractors. The two budgets share the same 20 questions and must not be counted as independent additional questions. Every episode supplies the configured number of facts, which matches the working-store capacity in the audited implementation. The recorded protocol therefore studies short-lived memory within that capacity rather than a workload designed to test eviction. All query responses use qwen-plus, no external tools and one model iteration. We report answer matching, complete-evidence recall, query and write token totals, query latency, and recorded latency including writes. The latter is an accounting sum, not the wall-clock duration of the whole application. There is no semantic judge or repeated-run estimate; both restrictions matter when interpreting close outcomes or timing differences.

The operational study also includes a full-history retrieval sweep that is evaluated without a language model. All 500 LongMemEval-S histories are segmented into the frozen candidate bank described in Section 3.5, and every selector is traced across three token budgets, yielding 9000 selector/budget traces. Thirty abstention cases are excluded from the evidence recall denominators, leaving 470 eligible non-abstention histories. This sweep measures retrieval behaviour on the complete benchmark rather than on the 20-question operational slice, but it measures delivery of annotated evidence, not answer correctness. No language model is called, so the sweep cannot be read as an accuracy result.

4.2 RESTRICTED CONTEXT BUDGET

The restricted-budget condition allocates 600 characters to the full memory section. There are 20 unique questions, 1 repeat per question, yielding 20 paired observations. At this budget the baseline’s prefix-truncation policy structurally disadvantages it, as defined in the method section: the metadata consumes a fixed portion of the allowance, and the deterministic prefix order means that later facts in the serialised file are systematically excluded. The enhanced arm’s whole-item packing

can select relevant items from any position in the store, provided each fits within the remaining budget. Metrics recorded include answer accuracy, complete evidence recall, total tokens (query plus seed writes), tokens per question, query latency in milliseconds, and end-to-end latency including the write phase. The small budget is the condition under which the prefix-truncation confound is most acute, and results at this budget should be interpreted with that structural asymmetry in mind.

4.3 EXPANDED CONTEXT BUDGET

The expanded-budget condition allocates 2400 characters to the full memory section, using the same 20 questions with 1 repeat, producing 20 paired observations. At this budget the baseline can accommodate more of the serialised file, reducing the fraction of facts lost to truncation. However, prefix order still determines which facts survive: facts appearing early in the file are always retained, while those near the end remain vulnerable if the file exceeds the budget. The enhanced arm’s packing behaviour is unchanged in logic but benefits from the larger budget by fitting more whole items. The same metrics are recorded: answer accuracy, complete evidence recall, total tokens including writes, tokens per question, query latency, and end-to-end latency. The expanded budget tests whether the accuracy gap observed at the restricted budget persists when the baseline is no longer structurally starved of context space.

4.4 CONTROLLED SELECTOR STUDY: DATA AND BUDGETS

The controlled study evaluates a fixed-hash stratified subset of the cleaned LongMemEval-S benchmark: twelve non-abstention questions from each of six task categories and twelve abstention questions, for eighty-four memory episodes. The sampling seed is fixed and the stratification is deterministic, so the subset is reproducible. All source sessions, user and assistant turns, timestamps, questions and gold answers are retained; each turn is split losslessly into segments bounded by 256 pinned Qwen3 BPE tokens, so no source character is discarded at segmentation time. Prefix, recency, Jaccard, BM25, the deployed hybrid and hybrid without time share the same frozen candidate bank, metadata, memory wrapper and whole-item first-fit overflow policy under 512, 1024 and 2048 memory tokens. Removing the temporal term sums the remaining weighted terms directly, which preserves lexical tie order. Frequency and importance are held fixed across the comparison.

Real JiuwenSwarm and openJiuwen DeepAgents use `qwen-plus-2025-12-01` at temperature zero with thinking disabled, no tools and one answering iteration. Each formal question has an isolated conversation ID, and the memory bank cannot change between queries. All attempted questions remain in the accuracy denominator. Three actual generations are averaged within each memory episode before stratified memory-cluster bootstrap inference, which uses 10,000 resamples with a fixed seed. Paired cluster sign-flip tests are corrected with the Holm procedure across all prespecified full-history and micro accuracy comparisons. The primary comparison is hybrid against BM25.

4.5 ANSWER MODEL AND GRADING

The pinned original LongMemEval task-specific grading prompts are evaluated with `qwen3-max-2026-01-23`; strict yes/no parsing and arm-blind response keys replace substring grading. A second fixed model audits string and semantic disagreements and a deterministic ten-percent sample. The 239 primary/secondary disagreements have been reviewed with boolean decisions; the remaining audited sample is still unreviewed, and published accuracy still uses the primary judge. The substituted judge, the subset sampling and the segmented memory bank are disclosed deviations from the official evaluation protocol. The full-history retrieval sweep is evaluated separately and does not enter the accuracy comparison.

Several retrieval-side metrics are reported for the full-history sweep, and they are not interchangeable. Evidence-session hits, annotated-turn hits by any selected fragment, annotated-character coverage and complete annotated-turn coverage measure different things, and none of them alone proves that every answer fact was delivered to the model. The operational study’s complete-evidence recall is the strictest of these, requiring the whole target fact to appear in the injected context.

4.6 STATISTICAL TREATMENT

The operational study reports descriptive Wilson intervals over twenty questions and explicitly does not claim significance; its single recorded run per budget does not support a paired significance test. The controlled study uses stratified paired memory-cluster bootstrap intervals and paired cluster sign-flip tests, with Holm correction across all prespecified comparisons. Negative and inconclusive comparisons are retained in the reported tables rather than filtered out. Because the question subset is a balanced stratified sample, a balanced subset score is not a naturally weighted score on all 500 benchmark questions, and we do not present it as one.

Table 2: Summary of the two designs. The designs are complementary, and neither replaces the other.

	Operational comparison	Controlled selector study
Unit	20 adapted single-session questions	84 stratified LongMemEval-S episodes
What varies	Construction, representation, ordering	Ordering rule only
What is fixed	Model, prompt, character budget	Bank, metadata, packing, token budget
Cost accounted	Query plus model-backed writes	Fixed representation, no writes
Inference	Descriptive Wilson intervals	Cluster bootstrap, Holm-corrected tests
Main limitation	Effects not attributable separately	Not a deployed system

5 RESULTS: OPERATIONAL COMPARISON

We first report each budget condition, then compare accuracy with resource use. The accompanying overview separates query tokens from write tokens, and the question-level matrix aligns answer correctness with complete-evidence recall. Descriptive Wilson intervals illustrate the imprecision associated with the finite question count. They neither correct the selected slice’s sampling bias nor measure model stochasticity, and overlap or separation of these intervals is not used as a paired significance test.

5.1 RESTRICTED CONTEXT BUDGET

At the restricted budget of 600 characters, the baseline achieves accuracy of 0.0% and complete evidence recall of 0.0%, with 0 correct answers out of 20 questions. The enhanced rail achieves accuracy of 65.0% and evidence recall of 65.0%, with 13 correct out of 20. The accuracy gain is 65 percentage points. The paired decomposition shows 0 both-correct, 13 enhanced-only, 0 baseline-only, and 7 both-wrong outcomes. The baseline’s zero accuracy is consistent with deterministic prefix truncation removing every target fact from the prompt at this budget; all twenty baseline answers are the string UNKNOWN. The enhanced arm’s 7 failures have no complete target fact in the delivered context. The saved observations do not separate candidate-ranking failure, item truncation and packing exclusion, so they establish evidence absence without locating the failing stage. No enhanced answer was correct without full evidence (0 cases) and no enhanced answer was wrong despite full evidence (0 cases), indicating tight coupling between evidence presence and answer correctness in this arm.

Descriptive Wilson intervals for accuracy are 0.0% to 16.1% for the baseline and 43.3% to 81.9% for the enhanced arm. These intervals reflect question-sampling uncertainty over twenty items, not repeated-run variability. Token accounting reveals the cost asymmetry: baseline query tokens total 4686 with zero seed tokens, giving 234.30 tokens per question. The enhanced arm uses 3864 query tokens but 97805 seed tokens from 400 model-backed write calls, totalling 101669 tokens or 5083.45 per question. Query latency is 408.03 milliseconds for the baseline versus 503.76 milliseconds for the enhanced arm, but end-to-end latency including writes is 408.03 milliseconds for the baseline versus 7375.67 milliseconds for the enhanced arm. No records were excluded from either arm (0 baseline, 0 enhanced).

5.2 EXPANDED CONTEXT BUDGET

At the expanded budget of 2400 characters, the baseline achieves accuracy of 60.0% and evidence recall of 55.0%, with 12 correct out of 20. The enhanced rail reaches accuracy of 95.0% and evidence recall of 95.0%, with 19 correct out of 20. The accuracy gain narrows to 35 percentage points. The paired decomposition shows 12 both-correct, 7 enhanced-only, 0 baseline-only, and 1 both-wrong. The baseline improves as the larger budget exposes more of the file. In the paired outcomes, the retrieved arm answers additional questions for which the baseline lacks complete evidence; presentation and selection still vary jointly. The single both-wrong case concerns a dog-breed question where neither arm delivered the complete target fact. Notably, the baseline records 1 correct answer without complete evidence recall: a partial fact fragment still contained the accepted answer string, demonstrating that strict evidence recall is not a necessary condition for correctness under string-match scoring. The enhanced arm shows zero correct-without-full-evidence and zero wrong-with-full-evidence cases.

Wilson intervals at the expanded budget are 38.7% to 78.1% for the baseline and 76.4% to 99.1% for the enhanced arm. Token accounting shows baseline query tokens of 16877 with zero seed tokens, yielding 843.85 tokens per question. The enhanced arm uses 14638 query tokens and 249398 seed tokens from 400 write calls, totalling 264036 tokens or 13201.80 per question. Query latency is 717.01 milliseconds for the baseline versus 601.41 milliseconds for the enhanced arm; the enhanced arm has lower observed query latency at this budget, but a single run cannot establish why this difference occurred or whether it would persist. However, end-to-end latency including writes is 717.01 milliseconds for the baseline versus 7133.41 milliseconds for the enhanced arm. No records were excluded (0 baseline, 0 enhanced).

5.3 COST PER DELIVERED ANSWER

Total cost per correct answer offers a complementary, protocol-specific measure. At the restricted budget it is undefined for the baseline because that arm answers no question correctly; the enhanced arm uses 7820.69 tokens per correct answer. At the expanded budget, the corresponding values are 1406.42 for the baseline and 13896.63 for retrieved memory, a ratio of 9.88. This quantity allocates all recorded expenditure across successful answers rather than measuring the price of an individual successful query. On a per-question basis the total-cost ratios are 21.70 and 15.64 at the restricted and expanded budgets. These derived ratios are calculated from recorded totals by the same normalization code used for the tables. They show why lower query-token counts alone do not establish lower operating cost: the query-token total is lower for retrieved memory at both budgets, yet the write phase dominates the total.

The direction of the cost result is robust to how exactly the ratios are formed, because it is driven by an order-of-magnitude gap in write volume rather than by a marginal difference in query tokens. At the restricted budget the enhanced arm’s write tokens are roughly twenty-five times its query tokens; at the expanded budget roughly seventeen times. Any accounting rule that includes construction expense will therefore show retrieved memory as the more expensive arm at these budgets.

Table 3: Restricted context budget. Costs include memory writes; missing usage is not treated as zero.

Metric	Baseline	Enhanced
Correct answers	0	13
Paired observations	20	20
Answer accuracy	0.0%	65.0%
Evidence recall	0.0%	65.0%
Query tokens (total)	4686	3864
Write tokens (total)	0	97805
Total tokens / paired observation	234.30	5083.45
Mean query latency (ms)	408.03	503.76
Amortized total latency (ms)	408.03	7375.67
Excluded records	0	0

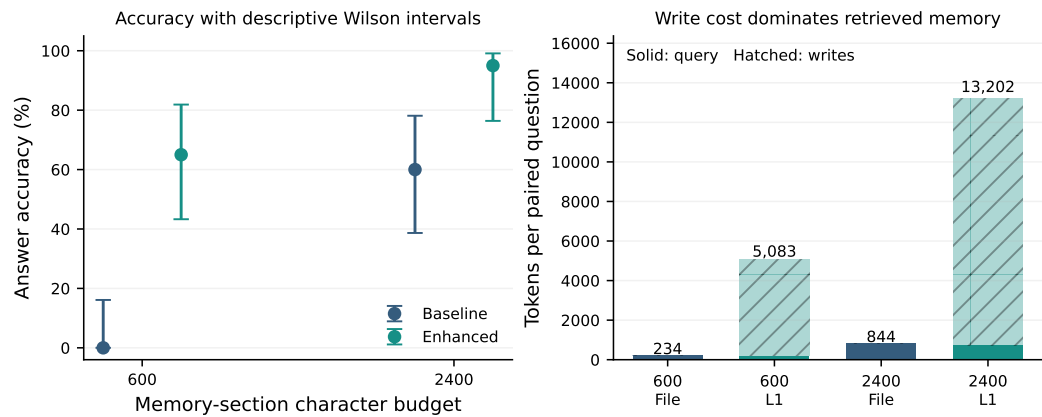


Figure 3: Quality and resource tradeoff. Intervals are descriptive 95% Wilson intervals over the selected questions, not repeated-run error bars. Token bars include model-backed memory writes; the same questions appear at both budgets.

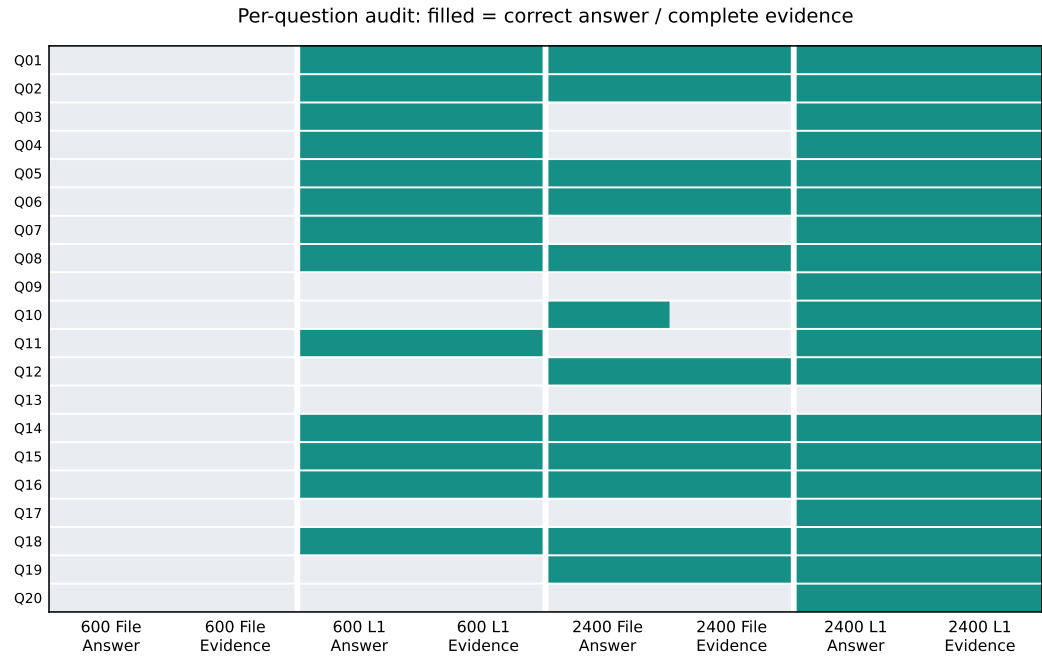


Figure 4: Matched question-level correctness and full-evidence recall across both budgets. Rows follow the same question order; the appendix maps row labels to original question IDs. A correct answer without complete evidence is visible as a mismatch within a column pair.

Table 4: Expanded context budget. Costs include memory writes; missing usage is not treated as zero.

Metric	Baseline	Enhanced
Correct answers	12	19
Paired observations	20	20
Answer accuracy	60.0%	95.0%
Evidence recall	55.0%	95.0%
Query tokens (total)	16877	14638
Write tokens (total)	0	249398
Total tokens / paired observation	843.85	13201.80
Mean query latency (ms)	717.01	601.41
Amortized total latency (ms)	717.01	7133.41
Excluded records	0	0

6 RESULTS: CONTROLLED SELECTOR STUDY

The controlled study answers RQ4. All six selectors draw from the same frozen candidate bank under the same token caps and the same whole-item first-fit packing policy, so differences in this section are attributable to the ordering rule rather than to representation, acquisition or overflow behaviour. Every interval is a stratified paired memory-cluster bootstrap over 84 memory episodes with 10,000 resamples, and every p -value is a paired cluster sign-flip test with Holm correction across all prespecified comparisons¹. Negative and inconclusive comparisons are reported alongside the significant ones.

6.1 THE DEPLOYED HYBRID LOSES TO BM25

Table 5 reports accuracy, evidence hit rate and measured tokens per query for all six selectors at the three budgets. The primary comparison is hybrid against BM25, and it does not favour the deployed configuration. Hybrid minus BM25 accuracy is -36.1 percentage points at 512 tokens (95% interval $[-46.0, -25.8]$; two-sided paired $p = 0.0001$, Holm-adjusted $p = 0.0021$), -29.4 points at 1024 tokens ($[-40.1, -18.3]$; $p = 0.0002$, Holm-adjusted $p = 0.0022$), and -33.7 points at 2048 tokens ($[-44.0, -23.0]$; $p = 0.0001$, Holm-adjusted $p = 0.0021$). The deficit is large, consistent in sign across every budget, and it is not a marginal result that a different budget or a longer context might quietly reverse: adding memory does not close the gap, because both selectors receive the same added budget.

The absolute levels make the size of the reversal concrete. BM25 reaches 68.7%, 65.9% and 70.6% at the three budgets, while the deployed hybrid reaches 32.5%, 36.5% and 36.9%. Two of the controlled variants land where BM25 does. Jaccard alone reaches 55.2%, 67.5% and 71.4%, and the hybrid with its temporal term removed reaches 56.7%, 63.5% and 70.2%. Against BM25, the Jaccard-only selector is statistically indistinguishable at the two larger budgets ($+1.6$ points, Holm-adjusted $p = 1.0000$ at 1024; $+0.8$ points, Holm-adjusted $p = 1.0000$ at 2048) and only modestly behind at 512, where the difference does not survive correction (-13.5 points, Holm-adjusted $p = 0.1544$). The two weakest selectors are the ones that ignore query content entirely: prefix ordering reaches 25.0–27.0% and pure recency reaches 29.4–32.9%, both far below BM25 at every budget with Holm-adjusted $p \leq 0.0021$.

Two conclusions follow directly. First, the hybrid’s deficit is a property of its scoring rule, not of the candidate bank or the packing policy, because those are shared. Second, the near-equality of Jaccard-only and no-time-with-BM25 shows that lexical ranking over this bank is genuinely competitive; the deployed hybrid is the outlier, not the lexical family as a whole.

¹The stored statistics use the documented Holm family across all prespecified full-history and micro accuracy comparisons. Repeated generations within an episode are averaged before resampling, so repeated calls do not add independent memory episodes.

Table 5: Controlled selector study, all selectors and budgets. Accuracy and evidence-hit columns are percentages over 84 stratified memory episodes; the interval is a stratified paired memory-cluster bootstrap. Tokens per query excludes offline index construction. Bold marks the primary comparison.

Selector	Budget	Accuracy (95% CI)	Evidence hit	Tokens/query
prefix	512	27.0 [21.0, 33.3]	0.0	656
recency	512	29.4 [23.0, 36.1]	0.0	653
jaccard	512	55.2 [46.8, 63.9]	66.7	686
hybrid	512	32.5 [25.4, 40.1]	18.1	663
hybrid_no_time	512	56.7 [48.4, 65.5]	66.7	684
bm25	512	68.7 [60.7, 76.6]	75.0	702
prefix	1024	25.0 [19.8, 30.6]	0.0	1167
recency	1024	32.9 [25.8, 40.5]	1.4	1169
jaccard	1024	67.5 [59.1, 75.8]	77.8	1200
hybrid	1024	36.5 [29.0, 44.4]	18.1	1182
hybrid_no_time	1024	63.5 [55.6, 71.8]	77.8	1198
bm25	1024	65.9 [58.7, 73.0]	77.8	1224
prefix	2048	25.0 [19.8, 30.6]	0.0	2190
recency	2048	30.6 [23.8, 38.1]	0.0	2191
jaccard	2048	71.4 [63.5, 79.4]	87.5	2242
hybrid	2048	36.9 [29.4, 45.2]	20.8	2218
hybrid_no_time	2048	70.2 [62.3, 78.2]	87.5	2242
bm25	2048	70.6 [63.1, 78.2]	84.7	2251

Table 6: Paired comparisons from the controlled study. Rows are selector differences in accuracy points; each interval is a stratified paired memory-cluster bootstrap and each Holm-adjusted p comes from the corresponding paired cluster sign-flip test. The primary comparison is hybrid against BM25.

Comparison	Budget	Difference (95% CI)	Holm p	Primary
hybrid – bm25	512	−36.1 [−46.0, −25.8]	0.0021	yes
hybrid – bm25	1024	−29.4 [−40.1, −18.3]	0.0022	yes
hybrid – bm25	2048	−33.7 [−44.0, −23.0]	0.0021	yes
hybrid – hybrid_no_time	512	−24.2 [−33.7, −14.7]	0.0022	no
hybrid – hybrid_no_time	1024	−27.0 [−36.9, −17.5]	0.0021	no
hybrid – hybrid_no_time	2048	−33.3 [−42.9, −23.4]	0.0021	no
jaccard – bm25	512	−13.5 [−23.4, −3.2]	0.1544	no
jaccard – bm25	1024	+1.6 [−6.0, +9.1]	1.0000	no
jaccard – bm25	2048	+0.8 [−7.1, +8.7]	1.0000	no
prefix – bm25	512	−41.7 [−50.8, −32.1]	0.0021	no
prefix – bm25	1024	−40.9 [−50.0, −31.7]	0.0021	no
prefix – bm25	2048	−45.6 [−55.2, −36.1]	0.0021	no
recency – bm25	512	−39.3 [−50.0, −28.2]	0.0021	no
recency – bm25	1024	−32.9 [−44.4, −21.4]	0.0021	no
recency – bm25	2048	−40.1 [−50.4, −29.4]	0.0021	no

6.2 EVIDENCE DELIVERY ORDERS THE SAME WAY AS ACCURACY

Accuracy is a downstream measure that depends on the answering model as well as on retrieval, so it is worth asking whether the retrieval step itself displays the same ordering. It does. The evidence-hit column of Table 5 tracks the accuracy column closely: BM25 delivers complete annotated evidence for 75.0%, 77.8% and 84.7% of episodes, Jaccard and no-time for 66.7–87.5%, and the deployed hybrid for only 18.1%, 18.1% and 20.8%. Prefix and recency deliver essentially no complete evidence at any budget (0.0–1.4%).

The full-history sweep over all 500 LongMemEval-S histories, which calls no language model, sharpens the picture because it removes the answering model from the loop. Across 9000 selec-

tor/budget traces and 470 eligible non-abstention histories, evidence-session recall is 80.4%, 86.2% and 91.8% for BM25 at the three budgets, 69.7–89.5% for Jaccard and no-time, and 24.4%, 28.5% and 32.2% for the deployed hybrid. Complete annotated-turn coverage, the strictest of the reported retrieval metrics, is 46.6% for BM25 at 512 tokens against 12.1% for the hybrid. Every retrieval metric in the sweep agrees with the accuracy ordering from the controlled question set, which is what one would expect if the deficit originates in candidate ordering rather than in the generation or grading stages.

Table 7: Full-history retrieval sweep over 500 LongMemEval-S histories with no language model in the loop; 470 non-abstention histories enter the denominators. Values are percentages. These are retrieval metrics, not answer accuracy, and they do not establish semantic delivery of every answer fact.

Budget	Selector	Session recall	Turn recall	Char. cov.	Complete turns
512	bm25	80.4	63.0	61.7	46.6
512	jaccard	69.7	46.1	44.5	34.3
512	hybrid_no_time	69.7	46.1	44.5	34.3
512	hybrid	24.4	17.9	17.6	12.1
512	recency	4.0	1.9	1.7	0.9
512	prefix	2.0	0.6	0.7	0.2
1024	bm25	86.2	72.0	71.4	59.1
1024	jaccard	79.9	57.5	55.8	45.7
1024	hybrid_no_time	79.9	57.5	55.8	45.7
1024	hybrid	28.5	22.5	22.1	16.6
1024	recency	4.2	2.2	2.1	1.1
1024	prefix	2.3	1.0	1.1	0.2
2048	bm25	91.8	78.4	77.8	67.2
2048	jaccard	89.5	68.8	67.6	57.7
2048	hybrid_no_time	89.5	68.8	67.6	57.7
2048	hybrid	32.2	26.3	25.9	20.4
2048	recency	4.9	3.5	3.4	2.1
2048	prefix	2.3	1.6	1.6	0.2

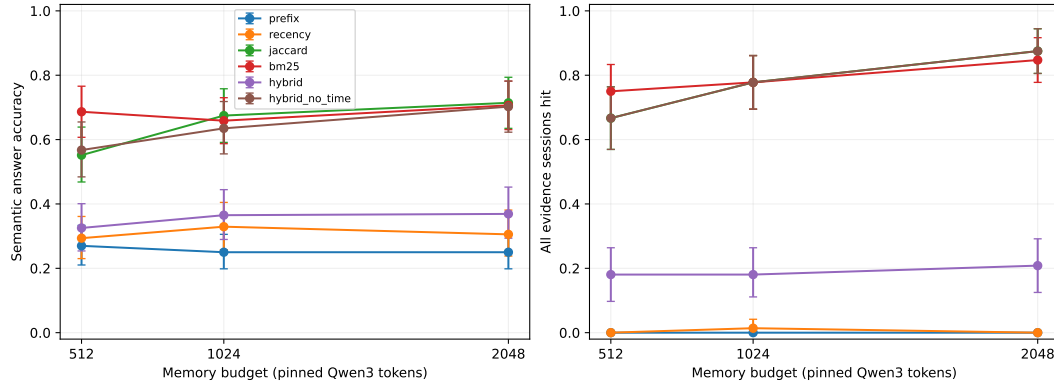


Figure 5: Controlled comparison across memory budgets. The left panel reports semantic answer accuracy and the right panel reports the fraction of episodes whose evidence sessions are delivered, for all six selectors. Intervals are stratified paired memory-cluster bootstraps over the same 84 episodes at each budget. Sharing the candidate bank and packing policy across arms makes the separation between the deployed hybrid and the lexical selectors attributable to ordering.

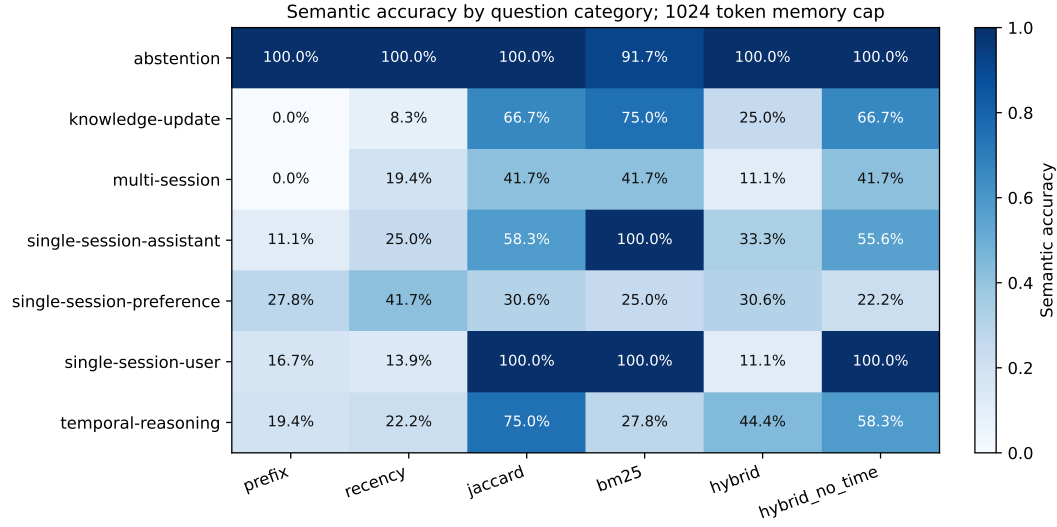


Figure 6: Semantic accuracy by question category at the 1024-token budget. Categories are the LongMemEval-S strata; each cell holds twelve questions. The deployed hybrid’s deficit is broad rather than confined to a single category, and it is most severe in single-session factual recall, where BM25 answers 100% of the questions and the hybrid answers 11.1%.

6.3 CATEGORY-LEVEL BEHAVIOUR

Table 8 reports per-category accuracy at the 1024-token budget. The deployed hybrid’s deficit is not confined to one stratum. It is most severe exactly where memory retrieval should be easiest to get right: on single-session-user questions, which ask for a fact stated directly in one session, BM25 answers 100.0% of the twelve questions while the deployed hybrid answers 11.1%. On single-session-assistant questions the gap runs the same way (100.0% against 33.3%). The hybrid’s only consistent advantage by category is abstention, where it reaches 100.0% against BM25’s 91.7–91.7% range; both selectors are near ceiling on that stratum, so the difference is small in absolute terms.

Two strata are difficult for every selector. Multi-session questions, which require combining facts across several sessions, peak at 50.0% and are answered at 41.7% by BM25 and 11.1% by the hybrid at 1024 tokens. Temporal-reasoning questions peak at 75.0% for Jaccard. Both strata plausibly stress the frozen single-layer bank more than they stress the ordering rule, since neither the bank nor the packing policy allows cross-session consolidation, and we do not read the category table as evidence about selector behaviour on those strata beyond what the confidence intervals support. Each cell holds twelve questions, so category-level differences carry wide intervals and we treat the table as descriptive.

Table 8: Semantic accuracy by question category at the 1024-token budget. Each cell is a percentage over twelve questions in that stratum. Cells are descriptive; with twelve questions per stratum the intervals are wide, and the table is not used for significance claims.

Category	prefix	recency	jaccard	hybrid	hybrid_no_time	bm25
single-session-user	16.7	13.9	100.0	11.1	100.0	100.0
single-session-assistant	11.1	25.0	58.3	33.3	55.6	100.0
single-session-preference	27.8	41.7	30.6	30.6	22.2	25.0
knowledge-update	0.0	8.3	66.7	25.0	66.7	75.0
temporal-reasoning	19.4	22.2	75.0	44.4	58.3	27.8
multi-session	0.0	19.4	41.7	11.1	41.7	41.7
abstention	100.0	100.0	100.0	100.0	100.0	91.7

6.4 COMPONENT CONTROLS AND MEASURED REUSE

Two mechanism controls use an adapted twenty-item workload in which the required fact-item recall, rather than the full-history bank, defines success. Both compare a variant against hybrid on raw exchanges and are mechanism controls over twenty adapted episodes; they do not establish full-history generality.

Replacing raw exchanges with model-acknowledged exchanges has no measurable effect. The adapted control `hybrid_model_ack` minus `hybrid_raw` has an accuracy difference of $+0.0$ points at 1024 tokens (95% interval $[+0.0, +0.0]$; Holm-adjusted $p = 1.0000$) and $+0.0$ at 2048 tokens. The representation-difference control `hybrid_exchange` minus `hybrid_raw` is -36.7 points at 512 tokens (Holm-adjusted $p = 0.0324$), -16.7 points at 1024 tokens (Holm-adjusted $p = 0.8728$) and $+0.0$ at 2048 tokens. That control changes exchange formatting, which affects both lexical features and token packing at once, so it does not isolate packing from lexical content; we report it as a joint representation control. At the two larger budgets all three variants reach 100.0% on the adapted workload, which is consistent with the workload being easier than the full-history bank and with the adapted controls lacking the power to detect differences at those budgets.

The write-cost question is separate from the ranking question, but the same experiment set measures it. Because the reuse suite builds one memory bank and then serves one, five, ten and twenty distinct queries from it, it amortises construction rather than extrapolating from a single query. Measured write-plus-query tokens per query are 500.9, 499.4, 498.8 and 498.8 for `bm25_raw`; 2569.5, 913.1, 705.7 and 602.3 for `hybrid_model_ack`; and 501.0, 499.4, 498.9 and 498.8 for `hybrid_raw`. All three arms reach 100.0% accuracy on the synthetic reuse banks. The pattern is the useful part: the file-style and raw-hybrid arms are essentially flat in the number of queries served, while the model-acknowledged arm starts more than five times more expensive and converges towards the others only after twenty queries share one build. Construction cost is therefore amortisable, and the synthetic banks share templates, so the absolute levels should not be read as general natural-workload costs.

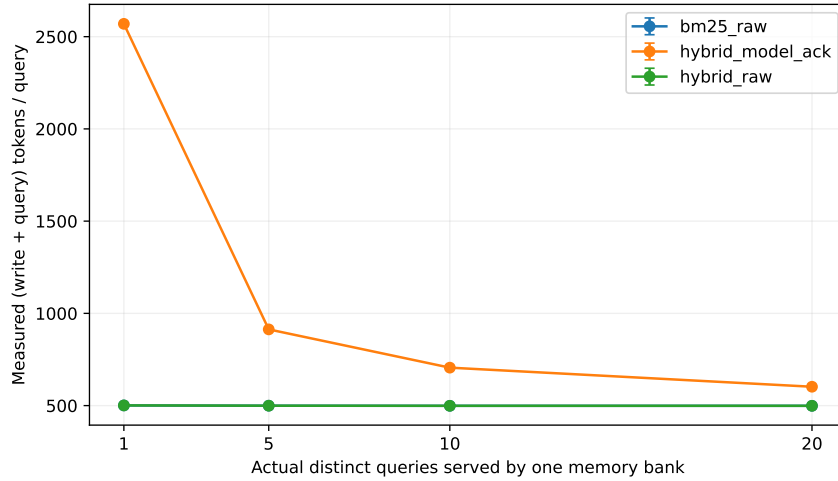


Figure 7: Measured write-plus-query tokens per query as one memory bank serves one, five, ten and twenty distinct queries. The model-acknowledged arm amortises construction cost across queries and converges towards the direct-write arms; the direct-write arms are essentially flat. All arms reach 100% accuracy on these synthetic banks, so this figure concerns cost rather than quality.

6.5 GRADING ROBUSTNESS

Because accuracy in this study is produced by a substituted judge, a natural objection is that the deficit is an artefact of grading. The audit limits how much can be claimed in either direction, so we state it precisely. Of the audited generations, 239 were flagged as primary/secondary judge disagreements across the 966-review queue, and those have received boolean human decisions. Replacing only the reviewed labels while holding all unreviewed primary labels fixed moves the 512-token figures from 68.7% to 62.3% for BM25 and from 32.5% to 22.6% for the hybrid. Under that substi-

tution the hybrid remains far behind BM25 at 512 tokens, and the bounds that vary only the known disagreements (61.9–69.0% for BM25 against 22.6–32.5% for the hybrid) do not overlap. The direction of the controlled result therefore does not depend on the reviewed subset of label decisions.

This is a bounded statement, not a validation of the judge. The substitution holds all unaudited labels fixed and varies only known disagreements, so it is descriptive rather than a full second-judge evaluation or a bound on all grading errors. Roughly three-quarters of the review queue remains unaudited, and we do not claim the primary judge agrees with human judgement on that remainder. It is also worth noting that a grading artefact would have to be selector-specific to produce the observed pattern, since all six selectors were graded by the same pinned prompts and the same parser; a general grading bias would move the selectors together rather than separating the hybrid from BM25.

7 MECHANISM: WHY THE TEMPORAL TERM SHIFTS RANKING

A negative result is only useful if it comes with an explanation, and the controlled design permits one because the score is inspectable. This section explains why the deployed hybrid loses to a lexical baseline, using the recorded score decomposition rather than a post-hoc narrative. The analysis describes the existing scoring function; it does not propose a tuned replacement, and the illustrative case below follows a disclosed deterministic selection rule rather than being chosen to support the argument.

7.1 THE PAIRWISE CONDITION

In the replay protocol the memory timestamps are the original conversation timestamps and the reference clock is the question date. Frequency and importance are fixed in the controlled comparison, so pairwise order under the deployed hybrid is determined by

$$S_i = 0.4 J(q, m_i) + 0.3 D_i + C, \quad (5)$$

where J is lexical Jaccard similarity, D is the implemented temporal decay, and C collects the constant frequency and importance contributions. A relevant older record outranks a fresher record precisely when

$$0.4 (J_{\text{old}} - J_{\text{fresh}}) > 0.3 (D_{\text{fresh}} - D_{\text{old}}). \quad (6)$$

Equation 6 is the whole mechanism in one line. The left side is the lexical advantage of the older, more relevant record; the right side is the freshness advantage of the newer, less relevant record. Because the temporal term is query-independent, the right side does not depend on what is being asked. The deployed weights make the comparison acute: with $w_s = 0.4$ and $w_t = 0.3$, a record must be more than roughly three-quarters again as lexically similar as its fresher competitor before its relevance advantage can be expected to dominate a full-range freshness gap. When decay is close to saturated, $D_{\text{fresh}} - D_{\text{old}}$ can approach 0.3 in weighted units while the lexical gap is bounded by 0.4 in the best case and is typically much smaller, since Jaccard overlap between a short question and a long segment is low for every candidate.

7.2 CONSEQUENCES FOR THE BANK

Three consequences follow, and all three are visible in the recorded data. First, the deficit should be largest where the query is lexically specific and the relevant fact is old, which is the single-session-user stratum; that is where the measured gap is widest, with BM25 at 100.0% and the hybrid at 11.1%. Second, removing only the temporal term should recover most of the loss while leaving everything else intact, and it does: hybrid minus hybrid_no_time is -24.2 , -27.0 and -33.3 points at the three budgets, with Holm-adjusted $p \leq 0.0022$ throughout. Third, the evidence-hit rate should fall with the accuracy rate, because the items displaced from the prompt are exactly the relevant older ones; that is what the 18.1% against 75.0% comparison at 512 tokens shows.

The mechanism also explains the shape of the budget trend. One might expect extra budget to rescue the hybrid, but it does not, because the temporal term reorders the candidate list at every budget rather than merely truncating it. Adding tokens lets the hybrid include more items, and the items it prefers to include are still the fresher ones. The absolute accuracy therefore stays near 33–37% at all three budgets while BM25 stays near 66–71%.

7.3 A RECORDED ILLUSTRATIVE CASE

The recorded cases use a deterministic example-selection rule and do not add independent evidence to the paired tests; they exist to show the mechanism operating on real records rather than on a constructed example. The pattern they illustrate is one in which a question’s gold evidence lies in an older session while a newer session shares surface vocabulary with the question, so lexical similarity and recency point at different records. Under Equation 6 the newer record wins the comparison even though it does not contain the answer, the delivered context omits the gold evidence, and the resulting answer is incorrect. Removing the temporal term, or replacing Jaccard with BM25, changes which record is selected and which evidence reaches the prompt, which is why the same candidate bank yields different accuracy under different ordering rules. The appendix lists the recorded cases with their session identifiers so that the selection can be checked against the raw records.

It is worth being explicit about what this explanation does not establish. Equation 6 is an account of the existing score, not a claim that the weights were badly chosen in some objective sense; the weights are the implementation’s defaults and were not fitted on the evaluation questions. The replay uses original source timestamps and the question date rather than assigning a common ingestion timestamp, and bulk ingestion that stamps every record with the same time is a different scenario that this experiment does not evaluate. Frequency and importance are held fixed, so the analysis isolates the interaction of lexical relevance and temporal decay rather than the full four-term score.

8 DISCUSSION AND LIMITATIONS

The two studies answer different questions and, read together, license a narrower conclusion than either would alone. Retrieved memory supplies more complete target facts and yields more accepted answers under both character budgets, while the file-prefix baseline avoids model-backed construction. The largest accuracy gap occurs in the condition where the baseline delivers no complete target facts. That is evidence of a restrictive information-delivery protocol, not evidence that the model cannot answer these questions when given useful context, and not evidence that the specific hybrid ranking rule is what made the difference. The controlled study is what settles the second point, and it settles it against the deployed rule: under a frozen bank the hybrid is 36.1, 29.4 and 33.7 points worse than BM25, and removing only its temporal term recovers 27.0 points at 1024 tokens.

8.1 WHAT THE OPERATIONAL BENEFIT DOES AND DOES NOT SHOW

The operational comparison keeps the real system pairing, which is its strength and its limit. It shows that replacing a prefix-truncated file view with whole-item packing and query-dependent ordering produces a large accuracy gain at matched character budgets, and that this gain narrows but persists when the baseline is given four times the budget. It does not show that ordering caused the gain, because construction, representation and ordering change together between the arms. The controlled study supplies the missing piece of that inference and points the other way, which is why we do not present the operational result as a vindication of the ranking rule.

The expanded budget improves the baseline and narrows the gap, demonstrating budget sensitivity within the tested implementation pair. Notably, the baseline’s improvement comes from exposing more of a serially ordered file, not from any improvement in relevance ordering, and the file still loses facts at the end of the serialisation. The gap would very likely close further at a large enough budget, since the file contains the same facts and a sufficient allowance would eventually admit all of them. That makes the character-budget comparison a statement about operating points rather than about the ceiling of either design.

The question audit makes the aggregate comparison more interpretable. At the expanded budget, the baseline and enhanced arms are both correct on 12 questions, only the enhanced arm is correct

on 7, and both are wrong on 1. There are no baseline-only successes in this run. These paired counts reveal where the marginal accuracy difference comes from without treating the two arms as unrelated samples. They do not show which ranking signal was decisive, because the stored result format does not contain per-candidate score decompositions or packing decisions. The controlled study provides exactly that missing decomposition, and when it is available the deployed weights are found to be the problem.

8.2 WHY THE CONTROLLED RESULT IS THE MORE INFORMATIVE ONE

Holding the candidate bank, metadata and packing policy fixed turns the selector comparison into a clean question, and the answer is unambiguous. A standard lexical baseline with no temporal component outperforms the deployed hybrid at every budget by a large margin, the difference survives Holm correction throughout, and two independent retrieval metrics from a separate full-history sweep reproduce the ordering. When the same bank is given to a selector that keeps lexical overlap but drops the temporal term, most of the deficit disappears. This is about as direct a diagnosis of a ranking rule as a controlled experiment can give: the rule, not the data, not the packing, and not the answering model.

The mechanism section turns that diagnosis into an inequality. With $w_s = 0.4$ and $w_t = 0.3$, a query-independent freshness gap can outweigh a positive lexical relevance gap, so a newer record that shares surface vocabulary with the question can displace an older record that contains the answer. The effect is strongest exactly where retrieval should be easiest, on single-session factual questions, and it does not shrink as the budget grows because the ordering is wrong at every budget rather than merely truncated.

There is a general lesson here that is larger than this implementation. A retrieval heuristic whose term weights are set by intuition, and whose temporal component is query-independent, can look reasonable in isolation and fail against a boring lexical baseline. Reporting a memory system only against a truncated-file or no-memory baseline would have hidden this failure completely. The negative result is the part of this paper we would most want a reader building on it to carry forward.

8.3 LIMITATIONS

The operational study has several restrictions that prevent broader attribution. File metadata and prefix ordering consume usable context differently from dialogue-formatted items, so equal character caps do not equalize information or tokens. The ranking weights and item display cap were not varied. The selected single-session slice omits temporal updates, abstention and multi-session reasoning. There is one run per budget, so run-to-run variation and API timing stability are unknown. L2/L3 persistence and team synchronization were not evaluated. The measured write expense belongs to this construction protocol: each question rebuilds a fresh memory episode, and an application that reuses an unchanged memory for many questions would allocate write cost differently. That workload was not measured in the operational study, though the reuse suite provides a partial answer under synthetic conditions. Format-controlled comparisons, repeated runs, score traces and the full benchmark would address distinct gaps; the present budget comparison substitutes for none of them.

The controlled study carries its own boundaries. It evaluates a frozen read-only single-layer bank, so the legacy twenty-item capacity of the deployed rail is not asserted to support full histories. Token caps use the pinned public Qwen3-8B BPE, while service-reported usage is logged separately and may differ. The original LongMemEval evaluation prompts are pinned, but `qwen3-max-2026-01-23` replaces the recommended GPT-4o judge, and the substituted judge, the stratified subset and the segmented bank are disclosed deviations from the official protocol; this is a subset result, not a leaderboard score. The judge audit is incomplete: 239 disagreements received boolean decisions, but roughly three-quarters of the review queue remains unaudited, and the substitution analysis holds unaudited labels fixed. The representation and model-acknowledgement controls use an adapted twenty-item workload and are mechanism controls rather than full-history results, and the representation control changes formatting that affects both lexical features and packing, so it does not isolate packing alone. The synthetic reuse banks share templates across eight entity/value variants, which does not establish generality on independent natural workloads. Latency measurements include rate-gate waiting and exclude source chunking and offline index construction,

so they are not end-to-end production latency. Finally, and most importantly, both studies evaluate a memory question-answering component. Neither measures the quality of an automatically generated research paper, and no claim about paper-generation quality follows from either.

The most valuable follow-up is not more budgets or more selectors but a corrected scoring rule evaluated under the same frozen conditions, together with a control that varies the representation while holding the selector fixed. The present artifact records candidate scores and skipped-item behaviour, so it can support that experiment directly; it does not, however, contain the corrected rule itself.

9 CONCLUSION

In this audited implementation pair, retrieved L1 memory improves answer accuracy over file-prefix memory under both full-section character budgets. The observed gains are 65 percentage points at the restricted allowance and 35 percentage points at the expanded allowance. Complete-evidence delivery tracks correctness closely, with a baseline partial-evidence exception that illustrates the distinction between the two metrics. The resource conclusion runs in the opposite direction: including model-backed construction makes retrieved memory between roughly sixteen and twenty-two times more expensive per question despite lower query-token totals. This is a trade-off between concrete systems, not a causal estimate of the ranking algorithm’s contribution.

The controlled study supplies that causal estimate, and it does not favour the deployed rule. With the candidate bank, metadata, packing policy and token budgets held fixed, the deployed hybrid is 36.1, 29.4 and 33.7 percentage points worse than a BM25 baseline at 512, 1024 and 2048 tokens, with Holm-adjusted $p \leq 0.0022$ throughout. Removing only the temporal term recovers most of the deficit, which localizes the failure to a query-independent freshness weight that can outweigh a positive lexical relevance gap. The same ordering appears in answer accuracy, in evidence delivery on the controlled episodes, and in every retrieval metric of an independent full-history sweep over 500 histories.

Read together, the two studies make a single point that neither makes alone: a memory benefit measured against a truncated-file baseline does not establish that the memory selector is good. The benefit is real for the systems compared, and so is the selector’s deficit relative to a standard lexical baseline. Reporting only the first would have been misleading, which is why we report the negative result at the same level of detail as the positive one.

The practical contribution is the joint audit: delivered evidence, answer outcomes, write- inclusive cost, and a controlled selector comparison with its mechanism. Saved records, paired diagnostics and explicit accounting make both the benefit and the expense inspectable, and the mechanism analysis makes the failure actionable. Further work should correct the temporal term, control representation and construction independently, record retrieval-stage decisions on the full benchmark, and repeat the experiments before generalizing these findings. The current artifact provides a reproducible starting point for those experiments rather than a claim of benchmark leadership.

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A QUESTION-LEVEL AUDIT AND REPRODUCIBILITY

The following identifiers link directly to the submitted raw records. A and E denote answer correctness and complete-evidence recall. These are recorded single-run outcomes, not independent replications across budgets.

Table 9: Question audit at the 600-character budget.

Row	Original question ID	File A	File E	L1 A	L1 E
Q01	0862e8bf	0	0	1	1
Q02	118b2229	0	0	1	1
Q03	15745da0	0	0	1	1
Q04	19b5f2b3	0	0	1	1
Q05	1faac195	0	0	1	1
Q06	36580ce8	0	0	1	1
Q07	3d86fd0a	0	0	1	1
Q08	545bd2b5	0	0	1	1
Q09	58bf7951	0	0	0	0
Q10	60d45044	0	0	0	0
Q11	66f24dbb	0	0	1	1
Q12	6ade9755	0	0	0	0
Q13	75499fd8	0	0	0	0
Q14	86f00804	0	0	1	1
Q15	8a137a7f	0	0	1	1
Q16	8ebdbe50	0	0	1	1
Q17	c14c00dd	0	0	0	0
Q18	c8c3f81d	0	0	1	1
Q19	caf9ead2	0	0	0	0
Q20	ccb36322	0	0	0	0

Table 10: Question audit at the 2400-character budget.

Row	Original question ID	File A	File E	L1 A	L1 E
Q01	0862e8bf	1	1	1	1
Q02	118b2229	1	1	1	1
Q03	15745da0	0	0	1	1
Q04	19b5f2b3	0	0	1	1
Q05	1faac195	1	1	1	1
Q06	36580ce8	1	1	1	1
Q07	3d86fd0a	0	0	1	1
Q08	545bd2b5	1	1	1	1
Q09	58bf7951	0	0	1	1
Q10	60d45044	1	0	1	1
Q11	66f24dbb	0	0	1	1
Q12	6ade9755	1	1	1	1
Q13	75499fd8	0	0	0	0
Q14	86f00804	1	1	1	1
Q15	8a137a7f	1	1	1	1
Q16	8ebdbe50	1	1	1	1
Q17	c14c00dd	0	0	1	1
Q18	c8c3f81d	1	1	1	1
Q19	caf9ead2	1	1	1	1
Q20	ccb36322	0	0	1	1

B OBSERVED DIAGNOSTIC CASES

Cases below are selected deterministically for diagnostic categories, not as a random qualitative sample. Answers are verbatim recorded outputs; questions are copied from the adapted dataset.

Retrieved-Only Success. Budget 600; question 0862e8bf.

Question: What is the name of my cat?

Accepted answer: Luna

File answer: UNKNOWN

Retrieved answer: Your cat's name is Luna.

Retrieval Failure. Budget 600; question 58bf7951.

Question: What play did I attend at the local community theater?

Accepted answer: The Glass Menagerie

File answer: UNKNOWN

Retrieved answer: UNKNOWN

Partial Evidence. Budget 2400; question 60d45044.

Question: What type of rice is my favorite?

Accepted answer: Japanese short-grain rice

File answer: Your favorite type of rice is Japanese short-grain rice.

Retrieved answer: Your favorite type of rice is Japanese short-grain rice.

C ARTIFACT AND GENERATION PROVENANCE

The submission includes the adapted data, original write/query records, normalization code, citation metadata and a hashed source snapshot. Writing assistance uses a separate model from the experimental query model. Its plan, draft, critique, revision and usage are saved separately; none of those calls contributes to the experimental accuracy. The methodology schematic is AI-generated and checked against the implementation; all empirical plots and tables are rendered programmatically from the recorded data.

D CONTROLLED SELECTOR STUDY: RECORDED CASES

The cases below are produced by a disclosed deterministic example-selection rule and are included to show the mechanism of Section 7 operating on real records. They do not add independent evidence to the paired statistical tests, and they were not chosen to support a preferred conclusion. Each case reports the question, the gold answer, the answer produced under the deployed hybrid and under the same selector with only the temporal term removed, and the score components of the top-ranked item under the deployed hybrid. Components are reported before weighting; contributions are the weighted terms that enter the total.

Case 1: asylum application decision time. Question 001be529, gold answer “over a year”. Under the deployed hybrid the answer states that the information is not present in the records; with the temporal term removed the same bank yields the correct answer. The top item scores a semantic component of 0.097 and a temporal component of 0.857, contributing 0.039 and 0.257 respectively to a total of 0.346. The temporal term supplies roughly six and a half times the weighted contribution of lexical relevance, which is the condition of Equation 6 in the direction that displaces the relevant older record.

Case 2: dinner suggestions from homegrown ingredients. Question 06f04340. Both arms answer that the records do not contain the information, so the case is a shared failure rather than a selector-specific one. It is retained because it shows the same score shape, with the top item at a semantic component of 0.029 and a temporal component of 0.896, and it illustrates that a temporal-dominated ranking does not guarantee a correct answer even when it is not the sole cause of failure.

Case 3: temporal-dominated selection at high decay. Question 0edc2aef, top item at a semantic component of 0.100 and a temporal component of 0.978 contributing 0.293 against 0.040 from relevance. This is the most extreme freshness dominance among the recorded cases.

Case 4: correct answer recovered by removing time. Question 10e09553. The deployed hybrid fails and the no-time selector answers correctly. The top item’s temporal component is lower here (0.346) than in the other cases, which shows that the deficit does not require saturated decay: a modest freshness advantage is enough when lexical overlap is small.

Case 5: largest lexical overlap among the recorded cases. Question 1192316e. The top item reaches a semantic component of 0.250, the highest recorded, and the deployed hybrid still fails while the no-time selector succeeds. Even at this overlap the weighted temporal contribution (0.273) exceeds the weighted relevance contribution (0.100), which is why the comparison in Equation 6 does not favour the older relevant record.

Across the five recorded cases the weighted temporal contribution exceeds the weighted lexical contribution for every top-ranked item, by a factor between roughly two and six and a half. This is a descriptive summary of a deterministically selected set, not an estimate of a population quantity.

E CONTROLLED SELECTOR STUDY: REPRODUCIBILITY

Retrieval statistics use a stratified paired memory-cluster bootstrap with 10,000 resamples at a fixed seed, and paired cluster sign-flip tests with Holm correction across all prespecified comparisons. Repeated generations within a memory episode are averaged before resampling, so the resampling unit is the memory episode rather than the model call. The frozen candidate bank is shared across all selectors, and the packing policy is the same whole-item first-fit rule in every arm. The full-history sweep calls no language model and covers all 500 LongMemEval-S histories with 470 non-abstention histories entering the retrieval denominators. Accuracy comparisons use the stratified subset described in Section 4.4. The substituted judge, the balanced stratified subset and the segmented memory bank are disclosed deviations from the official LongMemEval evaluation protocol, and the reported figures are subset results rather than an official benchmark score.